

AI APPENDIX

AI Tools Used: Gemini/Claude for brainstorming visualization ideas, debugging layout issues, and refining UI/UX logic.

1. Representative Prompts

Prompt 1: "How to display multiple countries at once and their corresponding CO2 and fossil fuel share percentages effectively?"

Prompt 2: "The AI suggested this might be too cluttered, but I only want to compare 3 countries tops. How can I implement a clean comparison view for a limited selection?"

2. What was Accepted or Rejected

Accepted: The idea of using normalization (per capita) instead of absolute values to make the comparison between 3 countries fair. The AI also helped with the logic for cross-filtering so that selecting a country on the map updates the comparison panel.

Rejected: The AI initially suggested a complex Radar Chart or a Stacked Bar Chart for all 38 countries. We rejected this because it led to the spaghetti effect mentioned in Section 5 of the report. We insisted on a limited selection to maintain clarity and user focus, prioritizing a cleaner side-by-side bar layout.

3. Example of Weak AI Output and Correction

The Issue: When asked how to show multiple countries, the AI initially warned against it entirely, claiming it would be "too cluttered" regardless of the number of countries. It pushed for a single-country view only.

The Correction: We exercised human reasoning by identifying that the "clutter" issue only applies to large datasets. We overrode the AI's suggestion and decided to implement a Multi-Select limit.